PERFECT COMPLEXES ON A QUASI-COMPACT OPEN ARE SUMMANDS OF RESTRICTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: same shape as L48237 dominate-by-admissible-blowup (special case of a delivered general lemma); proof environment immediately follows the statement

Editorial note. The label lemma-direct-summand-of-a-restriction names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Lemma lemma-lift-map-from-perfect-complex-with-support (delivered, L48096): for X quasi-compact quasi-separated, W a quasi-compact open, and P perfect on W, P is a direct summand of the restriction of a perfect object of D(O_X)

Source reference. Original reference 09IM: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3814–3814. Prerequisite pointer invoked at 3887; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let W be a quasi-compact open subspace of X. Let P be a perfect object of $D(\mathcal{O}_W)$. Then P is a direct summand of the restriction of a perfect object of $D(\mathcal{O}_X)$.*

Proof. Special case of Lemma 09IM. □

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.